(allow argument based on photoelectric efficiency)

8	**(a)**	**(i)**	*either* number = $6.02 \times 10^{23} \times (\{2.65 \times 10^{-6}\}/234)$			
			or number = $(2.65 \times 10^{-9})/(234 \times 1.66 \times 10^{-27})$	C1	
			$= 6.82 \times 10^{15}$	A1	[2]
		(ii)	$A = \lambda N$	C1	
			$604 = \lambda \times 6.82 \times 10^{15}$		
			$\lambda = 8.86 \times 10^{-14}\ \text{s}^{-1}$	A1	[2]
		(iii)	$T_{½} = \ln 2/\lambda$		
			$= 7.82 \times 10^{12}$ s	C1	
			$= 2.48 \times 10^{5}$ years	A1	[2]
	(b)	half-life is (very) long (compared with time of counting)		B1	[1]
	(c)	there would be appreciable decay of source during the taking of measurements		B1	[1]